```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set(style="whitegrid")
```

```
df = pd.read_csv("large_customer_churn_dataset.csv")

print("Dataset Loaded Successfully")
```

```
Dataset Loaded Successfully
```

```
df.head()
```

	CustomerID	Age	Gender	Tenure	MonthlyCharges	ContractType	InternetService	PaymentMethod	City	Churn
0	1	56	Female	62	662	One Year	Fiber	Net Banking	Bangalore	No
1	2	69	Female	47	3862	Monthly	NaN	Credit Card	Chennai	Yes
2	3	46	Male	23	2159	Two Year	DSL	UPI	Mumbai	No
3	4	32	Male	38	3245	Two Year	DSL	Net Banking	Chennai	No
4	5	60	Male	64	4834	Monthly	DSL	UPI	Bangalore	Yes

Next steps: Generate code with df New interactive sheet

```
print("Shape:", df.shape)
print("\nColumns:", df.columns)
df.info()
```

```
Shape: (100000, 10)

Columns: Index(['CustomerID', 'Age', 'Gender', 'Tenure', 'MonthlyCharges',
       'ContractType', 'InternetService', 'PaymentMethod', 'City', 'Churn'],
      dtype='object')
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 100000 entries, 0 to 99999
Data columns (total 10 columns):
 #   Column           Non-Null Count   Dtype
---  ------           --------------   -----
 0   CustomerID       100000 non-null  int64
 1   Age              100000 non-null  int64
 2   Gender           100000 non-null  object
 3   Tenure           100000 non-null  int64
 4   MonthlyCharges   100000 non-null  int64
 5   ContractType     100000 non-null  object
 6   InternetService  66747 non-null   object
 7   PaymentMethod    100000 non-null  object
 8   City             100000 non-null  object
 9   Churn            100000 non-null  object
dtypes: int64(4), object(6)
memory usage: 7.6+ MB
```

```
df.isnull().sum()
```

	0
CustomerID	0
Age	0
Gender	0
Tenure	0
MonthlyCharges	0
ContractType	0
InternetService	33253
PaymentMethod	0
City	0
Churn	0

dtype: int64

```
df.describe()
```

	CustomerID	Age	Tenure	MonthlyCharges
count	100000.000000	100000.000000	100000.000000	100000.000000
mean	50000.500000	43.500270	36.008190	2646.194700
std	28867.657797	14.970933	20.475132	1355.169697
min	1.000000	18.000000	1.000000	300.000000
25%	25000.750000	31.000000	18.000000	1473.000000
50%	50000.500000	43.000000	36.000000	2641.000000
75%	75000.250000	56.000000	54.000000	3819.250000
max	100000.000000	69.000000	71.000000	4999.000000

```
df["Churn"].value_counts()
```

Churn	count
No	60060
Yes	39940

dtype: int64

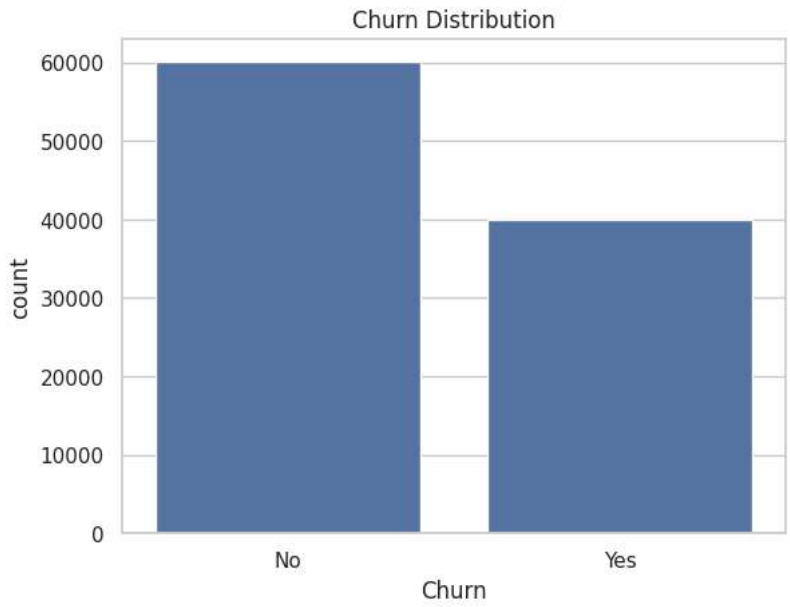
bold text Ask Business Questions (WRITE IN REPORT) Do high-paying customers churn more? Does tenure affect churn? Which contract type causes churn

```
df.groupby("Churn")["MonthlyCharges"].mean()
df.groupby("Churn")["Tenure"].mean()
```

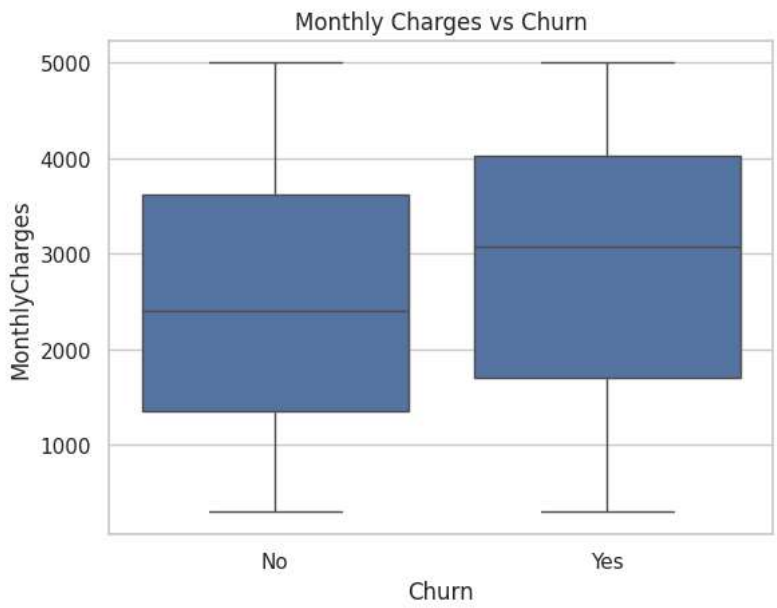
Churn	Tenure
No	39.038079
Yes	31.451978

dtype: float64

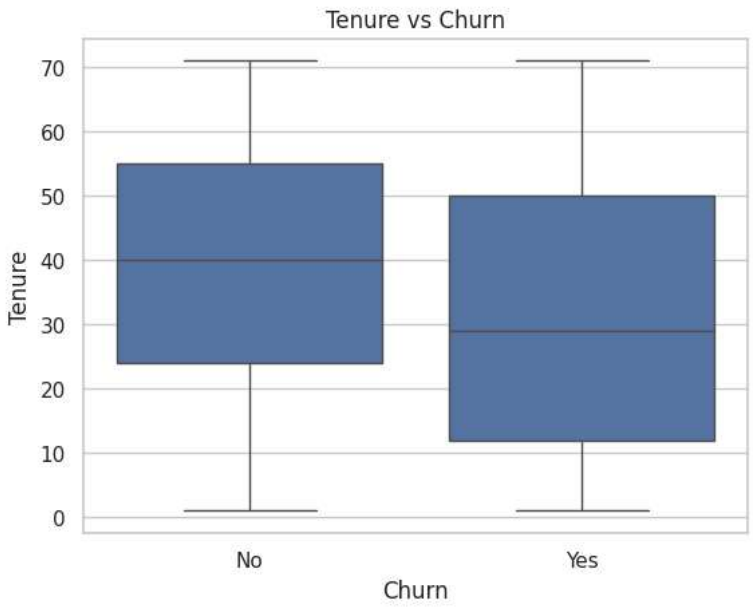
```
sns.countplot(x="Churn", data=df)
plt.title("Churn Distribution")
plt.show()
```



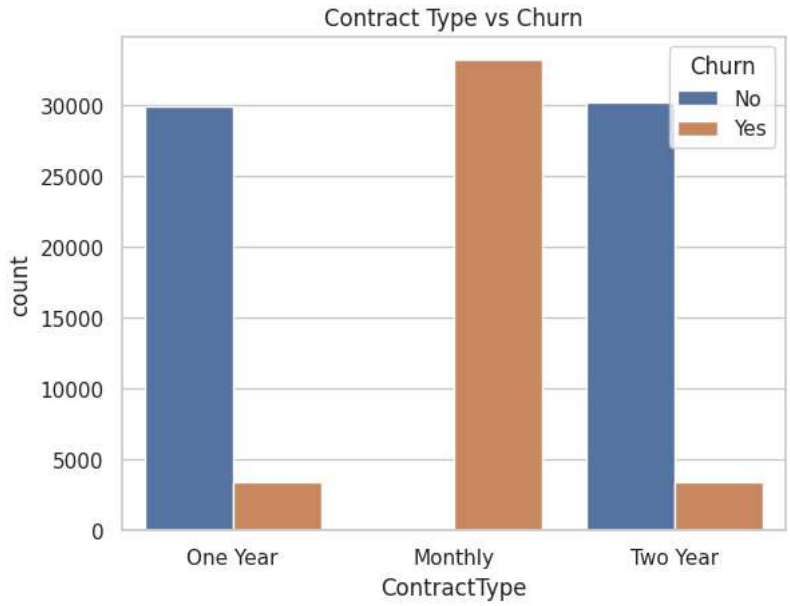

```
sns.boxplot(x="Churn", y="MonthlyCharges", data=df)
plt.title("Monthly Charges vs Churn")
plt.show()
```



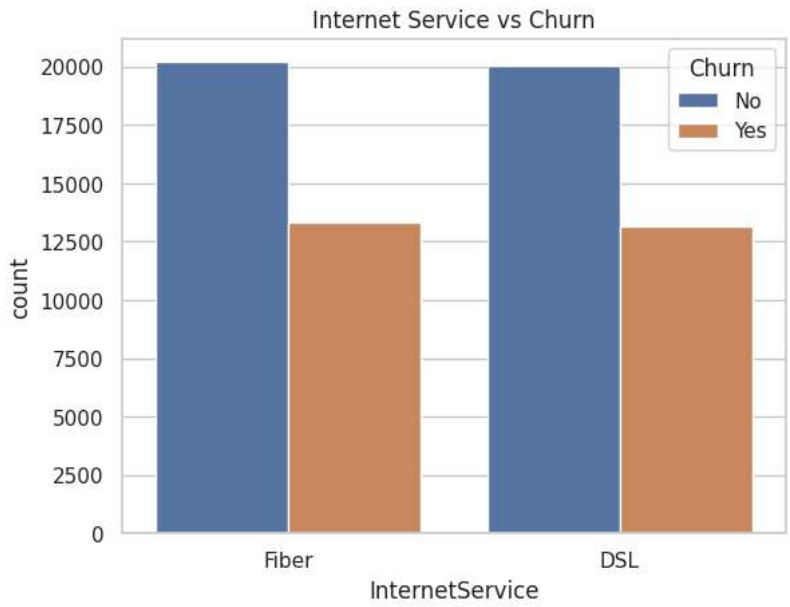

```
sns.boxplot(x="Churn", y="Tenure", data=df)
plt.title("Tenure vs Churn")
plt.show()
```



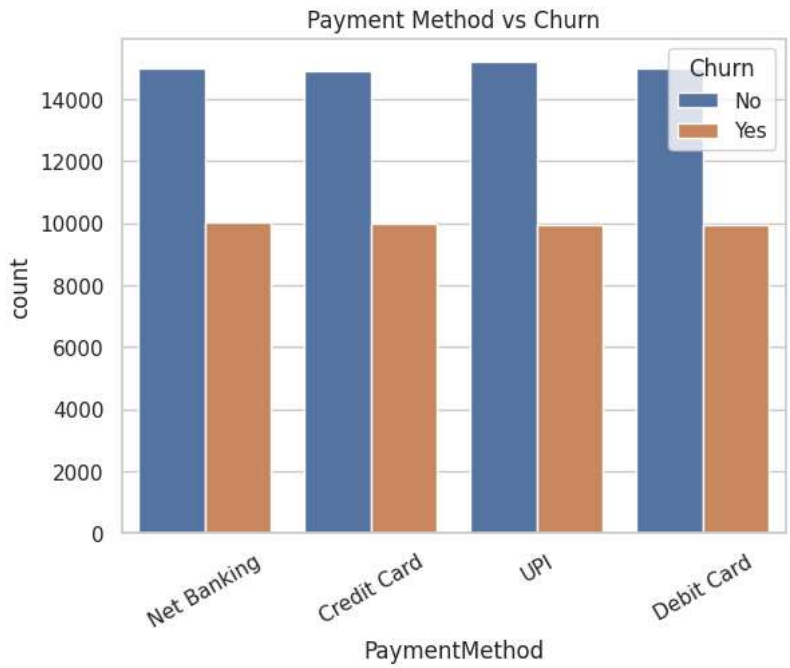

```
sns.countplot(x="ContractType", hue="Churn", data=df)
plt.title("Contract Type vs Churn")
plt.show()
```




```
sns.countplot(x="InternetService", hue="Churn", data=df)
plt.title("Internet Service vs Churn")
plt.show()
```

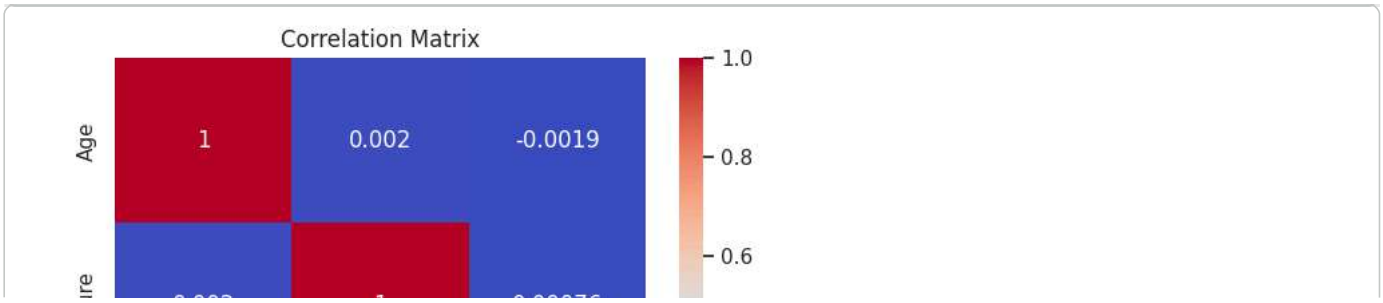



```
sns.countplot(x="PaymentMethod", hue="Churn", data=df)
plt.title("Payment Method vs Churn")
plt.xticks(rotation=30)
plt.show()
```




```
corr = df[["Age", "Tenure", "MonthlyCharges"]].corr()

sns.heatmap(corr, annot=True, cmap="coolwarm")
plt.title("Correlation Matrix")
plt.show()
```


Correlation Matrix
Age
1
0.002
-0.0019
1.0
0.8
0.6